

MINISTRY OF SCIENCE AND TECHNOLOGY
POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY



Assignment 06

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I. Task 1: Requirements Analysis

1. Actors

Customer – The main user who searches for products and orders and views returned results.

2. Functional Requirements

- Search products by keyword/text.
- Search products by voice.
- Search products by image.
- Search orders.
- View product information.
- View order information.

3. Non-Functional Requirements

- Usability: The system should be easy to use.
- Performance: Search results should be returned quickly.
- Maintainability: Components should be easy to modify and maintain.
- Extensibility: New search methods or AI models can be added.
- Modularity: UI, application logic, and data access should be separated.

4. Input Modalities

Mode	Input	Processing
Text	Keyword or natural-language query	Text search
Voice	Voice/audio query	Speech-to-text → text search
Image	Product image	Image similarity search

5. Search Outputs

- Product Search: A list of relevant products with product information.
- Order Search: Order information such as order ID, status, and total amount.

II. Task 2: Use Case Diagram

1. Actor: Customer: The main user who interacts with the system to search for products and orders and view related information.

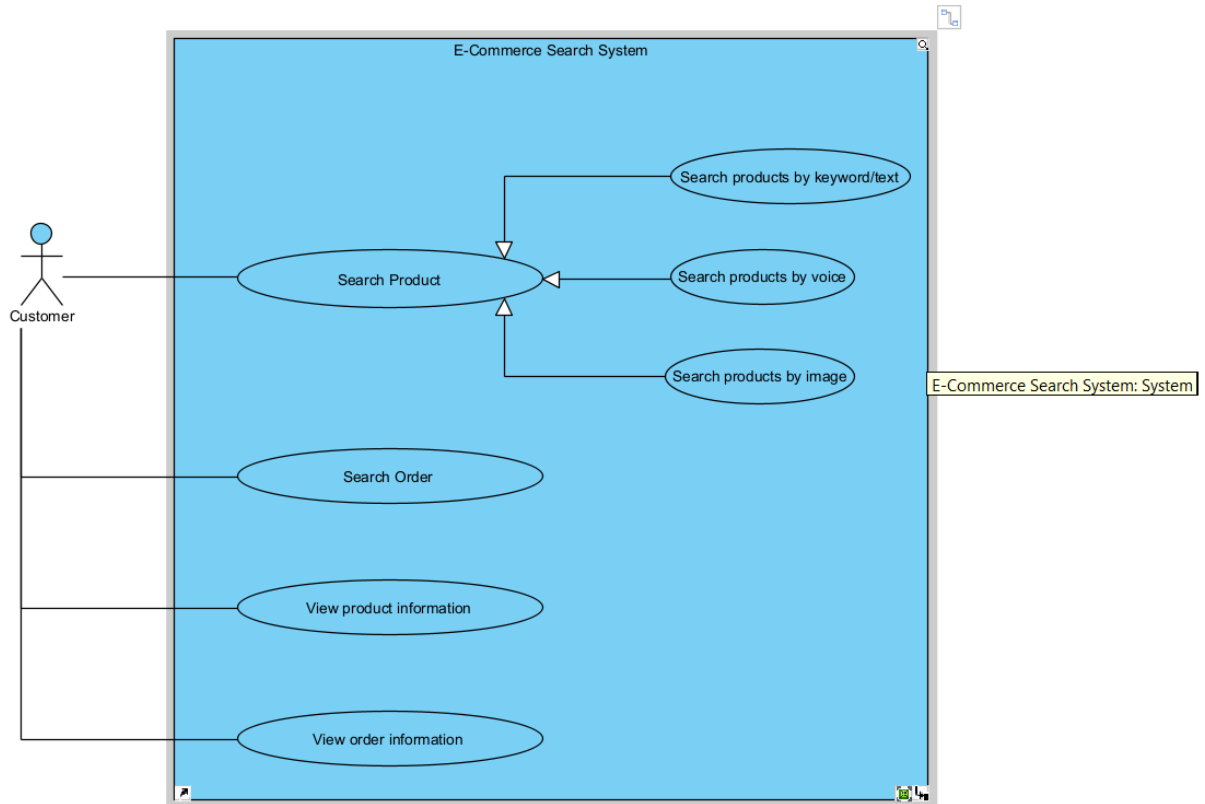
2. Main Use Cases

- Search Product: Allows the customer to search for products.
- Search products by keyword/text: Searches for products using keywords or natural-language text.
- Search products by voice: Searches for products using voice input.
- Search products by image: Searches for visually similar products using an image.
- Search Order: Allows the customer to search for order information.
- View product information: Allows the customer to view product details.
- View order information: Allows the customer to view order details.

3. Relationships

- The Customer has direct associations with:
 - Customer → Search Product

- Customer → Search Order
- Customer → View product information
- Customer → View order information
- Search Product is a general use case with three specialized search methods:
 - Search products by keyword/text → Search Product
 - Search products by voice → Search Product
 - Search products by image → Search Product



III. Task 3: Three-Layer Architecture

The system is organized into three layers: Presentation Layer, Application / Intelligence Layer, and Data Layer. Each layer has a separate responsibility, and the dependencies follow the direction:

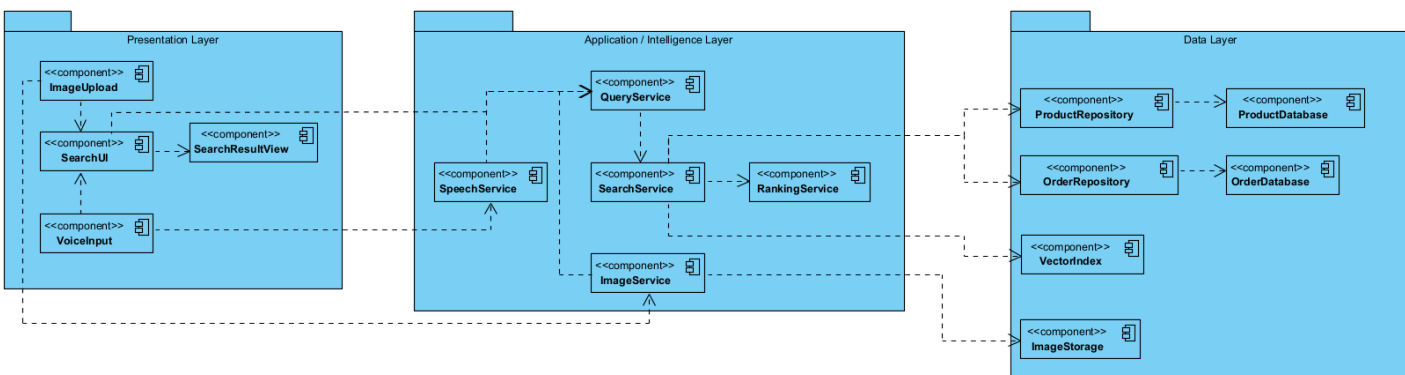
Presentation Layer → Application / Intelligence Layer → Data Layer.

1. **Presentation Layer**: contains SearchUI, VoicelInput, ImageUpload, and SearchResultView. This layer receives user input and displays the returned search results.
2. **Application / Intelligence Layer**: contains QueryService, SpeechService, ImageService, SearchService, and RankingService. This layer processes different input types, creates a common query representation, retrieves relevant data, and ranks the results.
3. **Data Layer**: contains ProductRepository, OrderRepository, VectorIndex, ProductDatabase, OrderDatabase, and ImageStorage. This layer stores and provides access to product, order, image, and vector data.

4. Main Interaction Flows

- **Text Search Flow**

- SearchUI → QueryService → SearchService → ProductRepository → ProductDatabase
- The user enters a text query through SearchUI. QueryService creates the query representation, and SearchService retrieves matching products from ProductRepository.
- **Voice Search Flow**
 - VoiceInput → SpeechService → QueryService → SearchService → ProductRepository
 - VoiceInput receives the user's voice input. SpeechService converts it into text, then QueryService creates the query representation and sends it to SearchService. This follows the voice-search process described in the assignment.
- **Image Search Flow**
 - ImageUpload → ImageService → QueryService → SearchService → VectorIndex
 - The user uploads an image through ImageUpload. ImageService processes the image, while QueryService creates the image-query representation. SearchService then uses VectorIndex to retrieve similar products. ImageService can also access ImageStorage for image data.
- **Order Search Flow**
 - SearchUI → QueryService → SearchService → OrderRepository → OrderDatabase
 - The system processes the order query and retrieves order information through OrderRepository.
- **Ranking Flow**
 - SearchService → RankingService
 - After relevant candidates are retrieved, RankingService ranks them before the results are presented to the user. The assignment explicitly separates retrieval from ranking.



IV. Task 4: Sequence Diagram

The Sequence Diagram describes the interaction when a Customer searches for a product using voice input. The diagram contains the required participants: Customer, SearchUI, SpeechService, QueryService, SearchService, ProductRepository, and RankingService.

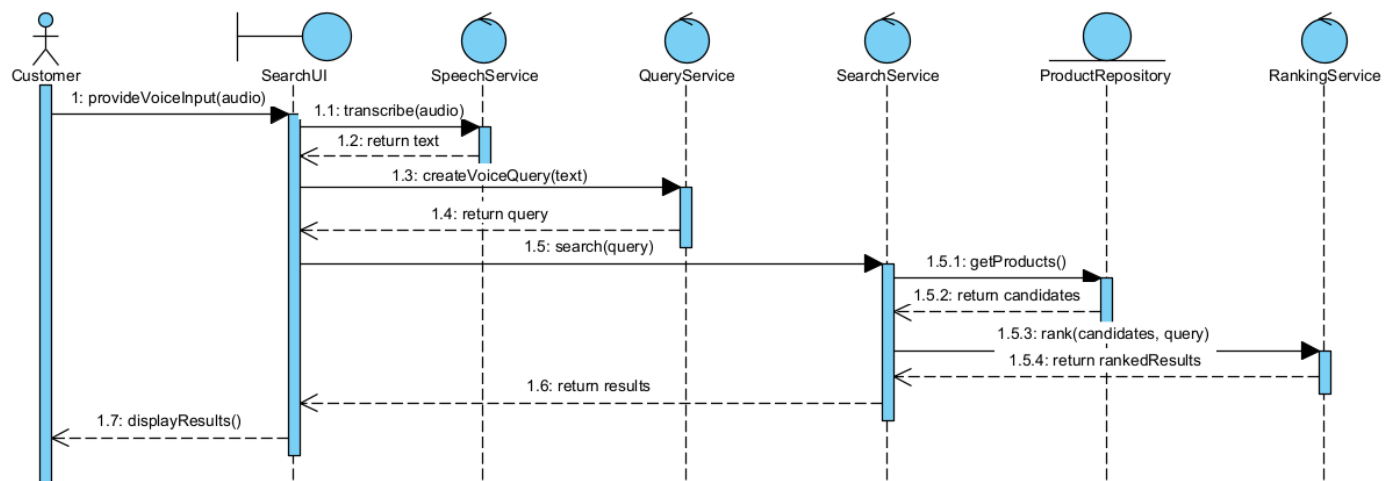
1. Participants

Participant	Responsibility
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Customer	Provides the voice query and receives the results.
SearchUI	Receives user input and displays search results.
SpeechService	Converts voice input into text.
QueryService	Creates a query representation from the transcribed text.
SearchService	Retrieves relevant products.
ProductRepository	Provides product data for the search process.
RankingService	Ranks the retrieved product candidates.

2. Interaction Flow

- 1. The Customer provides voice input to SearchUI.
- 2. SearchUI sends the audio input to SpeechService.
- 3. SpeechService converts the voice input into text and returns it.
- 4. SearchUI sends the text to QueryService to create a voice query representation.
- 5. QueryService returns the generated query.
- 6. SearchUI sends the query to SearchService.
- 7. SearchService retrieves products from ProductRepository.
- 8. ProductRepository returns the candidate products.
- 9. SearchService sends the candidates to RankingService.
- 10. RankingService ranks the candidates and returns the ranked results.
- 11. SearchService returns the results to SearchUI.
- 12. SearchUI displays the results to the Customer.



V. Task 5: Python Implementation

VI. Task 6: Demonstration